

STEM MAGAZINE 2026

ISSUE-1
COLOURS



GPS

Greet's Public School
A Project of Bethel Foundation

(Affiliated to the Central Board of Secondary Education)

Principal's Note

Dear Students, Parents, and Esteemed Members of the School Community, It is with great pride and enthusiasm that I present this edition of our STEM Magazine. In a world driven by innovation and rapid technological advancement, Science, Technology, Engineering, and Mathematics are not just subjects in a curriculum—they are powerful tools that shape the future.

At our institution, we strive to nurture curiosity, critical thinking, creativity, and collaboration among our learners. STEM education empowers students to question boldly, explore deeply, experiment fearlessly, and design solutions to real-world challenges. From robotics and coding to environmental research and mathematical modelling, our students continue to demonstrate exceptional talent and determination.

This edition of the magazine showcases remarkable projects, insightful research, and innovative ideas developed by our students and guided by our dedicated faculty. Each article reflects teamwork, resilience, and a commitment to excellence. I commend our teachers for fostering an environment where inquiry and innovation thrive, and I applaud our students for their passion and perseverance.

As we look towards the future, let us remain committed to embracing technology responsibly, promoting scientific literacy, and encouraging problem-solving that benefits humanity. The leaders, engineers, scientists, and innovators of tomorrow are being shaped in our classrooms today.

I encourage everyone to read this magazine with pride and inspiration. May it spark new ideas, ignite curiosity, and motivate us all to continue exploring the endless possibilities of STEM.

Warm regards,
Belinda Vivera

Principal Administration
Greets Public School



Secretary's Note

It is a matter of great pride to present this edition of the STEM Club Magazine of Greets Public School. The enthusiasm and effort shown by our students from Grades 7 and 8 truly reflect the spirit of innovation that STEM stands for. This issue covers a wide range of topics — from living fossils like the coelacanth and newly discovered species in Peru, to new findings about the core of Mars revealed through NASA's InSight mission. The features on octopuses, chameleons, dragonflies, and the return of otters highlight the diversity of life on our planet. The inspiring interview with Dr. T. Ashokan of Indian Institute of Technology Madras shows how curiosity and dedication can lead to meaningful work in robotics and innovation. I sincerely appreciate Sanjana Sreenath (VIII B), our Editor-in-Chief, along with contributors like Rudra Subramaniam (VII A), Jyothi Praveen (VII A), and every student who contributed to this publication. I would also like to acknowledge the continued support of the Management of Greets Public School in encouraging student-led initiatives such as this. I am equally grateful to our Club In-charge, Mrs. Rosemol T, for her valuable guidance and unwavering support in making this magazine a reality. What makes this magazine special is that it not only shares information but also encourages us to think deeper — especially about important questions such as whether we should try to bring back extinct species. As Secretary of the STEM Club, I am proud of the initiative and curiosity shown by our members, and I look forward to seeing this platform grow in the coming years.

- **Darin K Vimal**, XI B, Secretary, STEM Club.

Assistant Secretary's Note

As the Assistant Secretary of the STEM Club, I feel truly proud to present this magazine, which is a reflection of our team's creativity, curiosity, and dedication. Every page showcases the hard work, innovative ideas, and collaborative spirit of our members. This magazine is not just a collection of articles, but a celebration of scientific thinking and young minds eager to explore and discover. I sincerely appreciate everyone who contributed their time and effort to make this possible. Let this be a steppingstone towards many more achievements. Keep learning, experimenting, and inspiring the world around you!

- **Ajitesh S Nedungadi** XI B, Assistant Secretary, STEM Club

Editor in chief's note

STEM has always been something that interested me, the joy of your creations working perfectly after trying and failing over and over again, the way it explains our world and can be used to help people all over the world. I've always wanted to make such a magazine for kids with the same passion for discovering the wonders of our world as me and thanks to this great institution I can. As this is the first ever STEM magazine in our school, I am proud to publish this first issue and to say that I hope that this will be the start of an editorial team that is willing to make this magazine better than it is.

Sanjana Sreenath 8-B
Editor in Chief STEM Magazine

[NOTE:Those who have answers to the question on page:08 about extinct species please send replies in Word Document format to sanjanasreenathsoumya@gmail.com]

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STEM MAGAZINE

SCIENCE NEWS

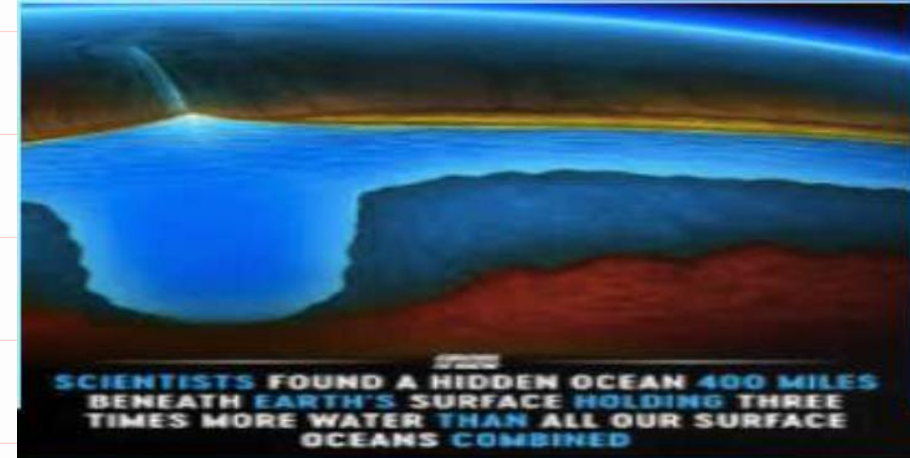
Living Fossil Spotted Again: The Indonesian Coelacanth. Scientists captured footage of a coelacanth—a fish thought extinct for 70 million years—swimming off Indonesia’s Maluku Islands. It’s a real-life "dinosaur fish".



Breakthrough against water scarcity: Scientists discovered a huge supply of fresh water hidden deep under the ocean floor. They believe it could stretch all the way from New Jersey to Maine.

It could help people in the future who don't have enough clean water. This water is under sand and rock- not floating like a bubble.

Scientists had to drill over 1,000 feet to reach it.



New creatures from Peru, an expedition in Peru's Alto May rainforest uncovered 27 brand new species, including 2.a "blob headed" fish, 3.a tree-climbing salamander and 1. a semi aquatic mouse right in a place bustling with human activity!



Woolly Mice: Mini Mammoths in the Making. Imagine a tiny mouse with shaggy fur, curly whiskers, and the ability to stay warm—scientists made that happen! These 'woolly mice' carry traits inspired by woolly mammoths and could teach us about cold resistance and genetics.

DO YOU THINK ITS RIGHT FOR US TO TRY TO BRING BACK EXTINCT SPECIES? SEND YOUR REPLIES IN WORD DOCUMENT TO THE EDITOR IN CHIEF

Scientists Have Bred Woolly Mice on Their Journey to Bring Back the Mammoth



The Return of Otters in Nepal (After 185 years, the Asia) small-clawed otter—the smallest otter in the world—was spotted again in Nepal's Makalu Barun National Park.



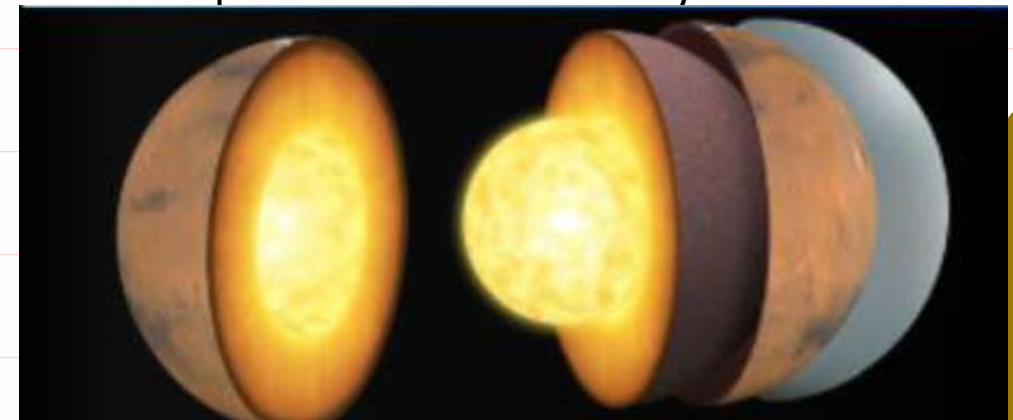
Bogs That Tell Tales: Ancient Megafauna DNA In peat bogs near Oslo, scientists discovered 75,000- year-old DNA from Ice Age giants like reindeer and musk ox. These genetic clues help us understand how animals survived harsh, ancient climates.



Sanjana Sreenath VIII-B

NASA's Insight Lander Reveals a surprise at The Very Core of Mars. Scientists using NASA's Insight seismic data have discovered that Mars has a solid inner core about 600 km wide, contradicting earlier beliefs that its core was entirely molten. Because Mars's iron core contains many light elements, scientists thought it was too hot to solidify.

This unexpected finding suggests Mars's interior and magnetic history work differently from Earth's, forcing a rethink of how the planet evolved internally.



Ameya D VII-B

Inspiring Minds in STEM:

An Interview with Dr. T. Ashokan, Professor at IIT Chennai by Sanjana Sreenath, 8 B, Greet's Public School, Kochi



Dr. T. Ashokan, Professor at IIT Chennai

I was honored to interview Dr. T Ashokan, a distinguished professor at IIT Chennai for the very first issue of our STEM magazine. His journey from a curious school student to a leader in robotics research is a reminder that passion, curiosity, and open-mindedness can take us far.

From Dreams of Science to a Career in Engineering:

Dr. Ashokan began his school years dreaming of becoming a scientist. Engineering wasn't widely spoken about back then and he mostly learned about scientists from textbooks. However, after completing his higher secondary studies, he discovered engineering through entrance exams—and his career took a new path. Although it wasn't a planned decision, he eventually found his true calling in academics: teaching, research, and mentoring young minds.

Building Robots and Solving Problems:

During his postdoctoral fellowship at Nanyang Technological University in Singapore, Dr. Ashokan worked on underwater robotics—an experience that gave him his first hands-on exposure to building real robots. Later, at IIT Chennai, he collaborated with DRDO (Defense Research and Development Organization) to develop an underwater robot. This marked the beginning of his long and impactful journey in robotics.

Over the years, he has contributed to spreading robotics research in India, helped set up student chapters, filed numerous patents, and published more than 200 papers.

Innovation for Society:

One of Dr. Ashokan's most inspiring experiences was the Stanford India Biodesign Fellowship, where he worked on developing medical devices suited for Indian hospitals. With his team, he created a device to support accident victims, which was later patented and commercialized. This experience inspired him to start a course on medical device innovation at IIT, encouraging students to design technologies that truly serve people.

Advice for Young students:

When asked what message he would like to give students exploring STEM careers, Dr. Ashokan emphasized:

- Keep an open mind. Interests may change as you grow—don't lock yourself into a rigid career path too early.
- Follow your passion. Don't choose a career because of peer pressure; excel in what you truly love.
- Stay curious. Always ask "Why is it done this way? Can it be done differently?"
- Develop problem-solving and hands-on skills. Projects, experiments, and tinkering help



As members of the STEM club and future innovators, we can take inspiration from Dr. Ashokan's journey and remind ourselves: the key to success is not following the crowd but following our curiosity with passion.

By : Sanjana Sreenath 8-B
(editor in chief)

THE WONDERS OF THE ANIMAL KINGDOM: OCTOPI

Octopi are soft-bodied, eight-limbed mollusks belonging to the order Octopod class Cephalopoda, which also includes squids and cuttlefish. With around 300 known species, a group of octopi is known as a consortium.

All octopi possess venom, though only a few are dangerous to humans. The greater blue-ringed octopus is among the most venomous animals on Earth—its tetrodotoxin can kill ten adult humans causing rapid respiratory failure.

Brains Without Backbones

Octopi are remarkably intelligent for invertebrates. Unlike humans, their intelligence evolved along a completely different evolutionary path, diverging about 2 billion years ago. Without a shell and living mostly solitary lives, octopi developed advanced survival skills like camouflage, tool use, and mimicry.

Octopus Species

Lets start with...

The Mimic Octopus



The best imitator, capable of mimicking up to 15 marine animals—including lionfish, jellyfish, and flatfish.

Scientific Name: *Thaumoctopus mimicus*

Size: 23 in (60 cm)

Giant Pacific Octopus



The largest known species, weighing up to 75 kg and stretching over 20 feet.

Scientific Name: *Enteroctopus dofleini*

Habitat: North Pacific

Known for using discarded coconut

Coconut Octopus



shells and human trash as tools. Even more amazing: it walks on two limbs, a rare example of bipedal movement in invertebrates.

Scientific Name: *Amphioxus marginatus*

Habitat: Pacific & Indian Oceans

Caribbean Reef Octopus



Famous for its vivid color and camouflage abilities, often seen in coral reefs and shallow waters.

Scientific Name: *Octopus briareids*

Habitat: Caribbean & Western Atlantic

Flapjack Octopus



The cutest octopus. It flattens itself to appear harmless, resembling a floating pancake.

Scientific Name: *Opisthoteuthis californiana*

Habitat: Eastern Pacific

Bimac Octopus



Nicknamed the “friendliest octopus”, this species is popular in the exotic pet trade due to its curious and interactive behavior (you should still not buy octopi as pets as they are currently all taken from their wild homes for the pet trade)

Scientific Name: *Octopus bimaculoides*

Habitat: Baja California

Octopi continue to amaze with their intelligence, adaptability, and otherworldly charm. Whether walking on two arms or blending into coral, they truly are one of nature's most fascinating creatures.

Sanjana Sreenath VIII-B

Creature Feature - The Colorful Chameleon

Chameleons are some of nature's most fascinating reptiles, best known for their magical ability to change color. But their color-changing powers are not just for blending into their surroundings—they serve many amazing purposes!



The Magic of Color: Chameleons use special skin cells called chromatophores, which contain different pigments. By expanding or shrinking these cells, they can create dazzling shades of green, yellow, red, blue, and even orange.

This process helps them regulate their body temperature, communicate with other chameleons, and protect themselves from danger.

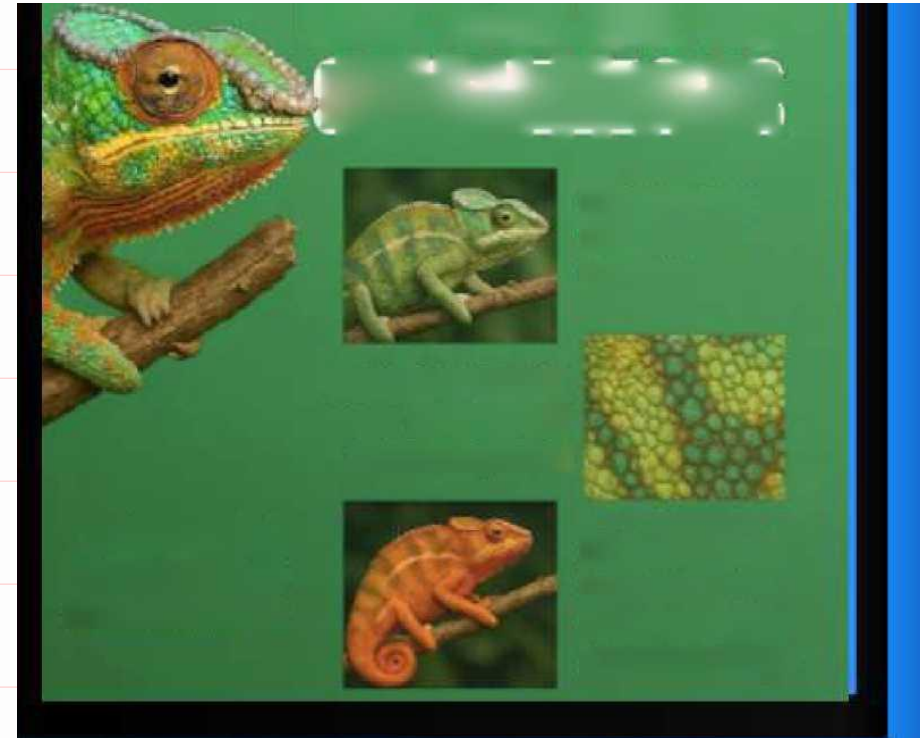
Why Do They Change Color?

Camouflage: To hide from predators

Communication: Bright colors may signal aggression or attract a mate.

Temperature Control: Darker colors absorb heat, while lighter shades help them stay cool.

The Colours Of Chameleons



A Living Rainbow

From the rainforest canopy to desert shrubs, chameleons are living rainbows, showcasing how science and nature work in harmony. Their unique skin is like a natural laboratory, combining physics, biology, and survival in one tiny creature.



Rudra Subramaniam, VII A



The Dragonfly (or "Thumbi" in Malayalam) can be seen everywhere across Kerala. Now don't let the name dragonfly fool you cause dragonflies are not actual flies they are in the scientific order Odonata and have two pairs of wings while flies don't. Dragonflies are super cool since they are agile fliers, they can fly backwards and can hover in the air, also say bye to the mosquitos when there is a dragonfly around cause dragonflies eat over a 100 mosquitos in a single day. A dragonflies life cycle consists of an egg, nymph, and adult.

A dragonfly adult



A dragonfly nymph



Jovin Sijoy VIII-C

Something Interesting in Green

- Rudra Subramaniam, VII A

1. The Planet's Favorite Colour, Around 70% of Earth's land is covered with green plants. That's why astronauts often call Earth the "green planet" when seen from space.

2. Green Blood?

Not just plants—some animals are green inside too! The New Guinea skink (a lizard) has green blood because of a chemical called biliverdin.



3. Some jellyfish and sea creatures glow with a green light in the dark oceans. This happens because of green fluorescent protein (GFP), which scientists now use in labs to study cells.

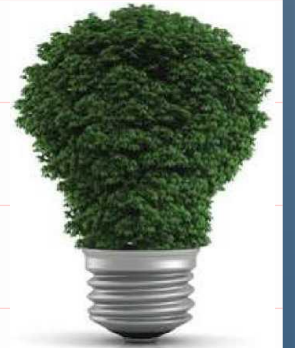


4. Superfood in Green, Green vegetables like spinach, broccoli, and kale are packed with iron, calcium, and vitamins. Eating green keeps our body strong and healthy—nature's free medicine!



5. Green Tech

The word "green" also means eco-friendly technology—like solar panels, wind turbines, and electric cars that protect our planet.



Wow Takeaway:

From glowing jellyfish to eco-friendly energy, green is more than a colour—it's a story of science, and the future!

SKY BLUE FACTS

- In heraldry, sky blue is called **blue celeste** and is rarer than standard blue.
- Ancient Egyptians created a pigment close to sky blue using **copper calcium silicate**, one of the first synthetic colors.



- Sky blue was once considered a **feminine color** in Europe, before pink took that role.
- In psychology, sky blue is linked more to **creativity and open-mindedness** than darker blues, which are tied to authority.
- Some cultures see sky blue as a **protective color**—the “evil eye” amulet is often made in that shade.



- The color **sky blue** was officially added to the English language in **1728**.
- In Argentina, sky blue is on the **national flag**, symbolizing the sky parting during independence.



- Astronauts say the thin **blue halo of Earth's atmosphere** looks exactly like a pale sky blue from space.



- In medieval art, sky blue pigments were so rare they were sometimes **reserved only for angels**.
- Some sports teams choose sky blue because it's linked with **fair play and honesty**.
- In color therapy, sky blue is believed to **lower blood pressure** more effectively than darker blues.

BLOOD PRESSURE CHART



- “Sky blue” in Japanese (空色, *sorairo*) is a distinct named shade, separate from normal blue (*ao*).
- Sky blue fabrics were once prized in trade routes because the dye was **harder to fix** than navy or indigo.



Jyothi Praveen VII-A



The Mirror Puzzle

Scenario:

Imagine you are standing in front of a large mirror holding a book in your hand.

The letters on the book appear flipped left to right, but not upside down.

Now, what if instead of a mirror on the wall, the mirror was placed flat on the floor beneath you?

Question:

How would your reflection look in this case? Would the letter still flip, or would something different happen?

Think About It

Mirrors don't actually "swap" left and right-they reverse front to back.

When placed on the floor, the mirror changes the up-down direction instead of left-right

The Answer:

If the mirror is on the floor, your reflection will appear upside down, but the text in your book will still be readable left to right!

STEM Connection:

This puzzle shows how light rays and reflections work. It also explains why periscopes, kaleidoscopes, and even telescopes use mirrors in clever ways to control what we see.

Challenge Extension

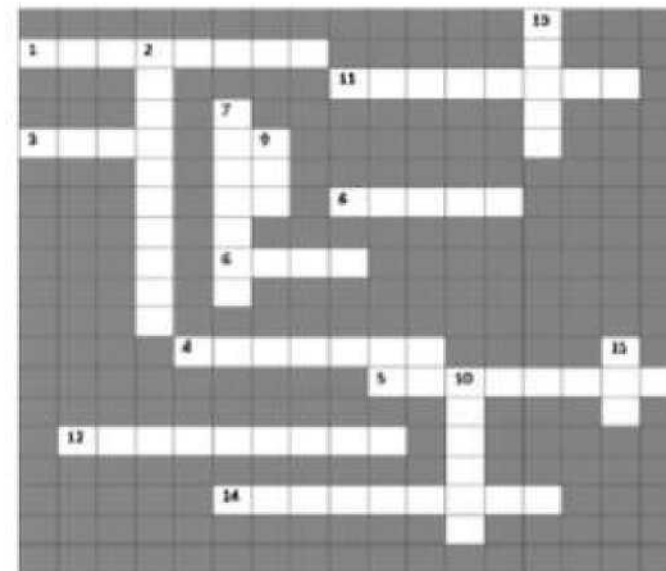
Try this at home with a small mirror-hold a paper with words and see how reflections change when you move the mirror to different positions (wall, floor, ceiling). Record your observations like a young scientist!

- | | | | |
|---------|----------|----------|----------|
| 1.blue | 4.red | 7.green | 10.grey |
| 2.pink | 5.purple | 8.yellow | 11.brown |
| 3.black | 6.orange | 9.white | |



Jyothi Praveen VII-A

IT DICTIONARY CROSSWORD



ACROSS

1. A computer program that translates source code into machine code.
3. Where is India's Super Computer of India PARAM located?
4. A specific instruction designed to do a task.
6. What are physical devices of a computer called?
8. Microsoft's web browser succeeding Internet Explorer.
11. A security system that prevents unauthorized access to network.
12. A well-known graphic design software by Adobe.
14. MS is the common abbreviation for _____.

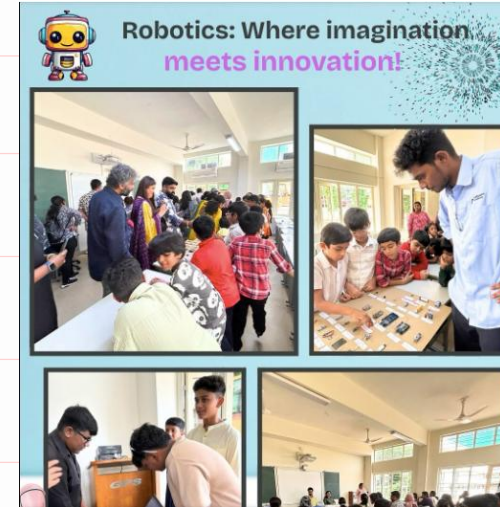
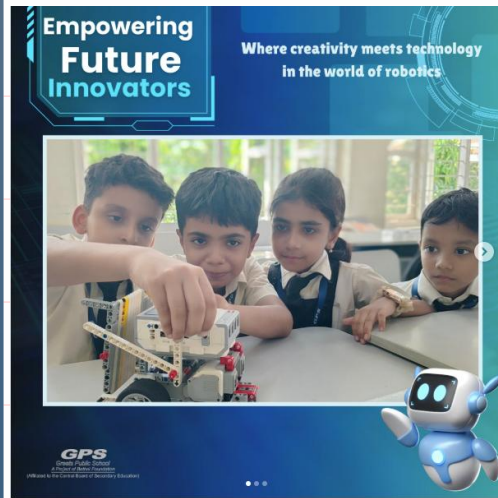
DOWN

2. A software utility used to create presentation with slides.
7. A computer program where you can look up an information on the internet.
9. A small software program that runs on a mobile device.
10. Booting a computer system again is called _____.
13. First neural network computer.
15. Which monitor looks like a television and are normally used with non-portable computer systems?

Aranya D VII B

Answers: 1. Compiler, 2. PowerPoint, 3. Pune, 4. Program, 5. Hardware, 6. Edge, 7. Browser, 8. Virus, 9. App, 10. Reboot, 11. Firewall, 12. Photoshop, 13. SNARC, 14. Microsoft, 15. CRT

Our school's finest moments in STEM:



Saral.

Multilingual · Simple · Saral

Five students of our very own school built a new business management app named SARAL. Rishi Rohith, Ayush Manoj, Neha R Shameesh, Esther Sony, and Poornathray Praveen designed SARAL for the Viksit Bharat Buildathon 2025. SARAL helps Indian entrepreneurs run daily operations. You open the application and select your preferred Indian language. You track sales and monitor inventory in real time. The platform works offline first. You keep your data on your device. You sync your data securely when you connect to the Internet.



scan to check us out!

You receive instant alerts when stock drops low. You avoid business delays because you restock early. You add a Google Gemini API key to activate AI features. The AI analyzes your revenue statistics. The AI gives you clear insights to improve your business decisions. You create individual accounts for your employees. You control employee access permissions. You export and import your business records to back up your data. You build structured business operations with SARAL. You support local growth by adopting tools built in India.

SARAL manages your inventory levels and tracks your sales records. You view monthly revenue reports on a clean dashboard. You manage employee profiles with specific access levels. The app stores all sensitive business information on your local hardware. You protect your privacy by keeping data off the cloud. You integrate AI features to automate repetitive tasks. You focus on expanding your business operations. You help India achieve a developed future.

THANK YOU